The Campus of Tomorrow

Higher Colleges of Technology

كليات التقنية العليا

CDS 2413- PROGRAMMING FOR DATA SCIENCE

Week 5

Week 5	CLO 2: Apply data analysis and visualization techniques to support data-driven decision-making

Document Revision CONTROL (DRC)

Version	Author	Effective Date	Change Description	DRC No
1.0	Mohammed Abdul Rahim	2/05/2023	Define the first version	001

X.Y
X: Major change
Y: minor change

Lecture Outline

- Correlation
- Types of Correlation
- Python Programming for Correlation
- Hypothesis Testing
- Python Programming for Hypothesis Testing

LECTURE OBJECTIVES

- At the conclusion of the lecture, the student should be able to perform the tasks below:
  - Understand the significance of variable correlation
  - Understand categories of correlation
  - Understand the concept of hypothesis test
  - Capable of examining the intent of Hypothesis Testing
  - Learn how to write Python code for correlation and hypothesis test

Correlation

Correlation(معامل الارتباط)

- **Correlation Coefficient (r)**:
- Measures the **relationship** between two variables.(يقيس العلاقة بين متغيرين)
- Ranges between **-1 and 1**.
- Indicates the **strength** and **direction** of the relationship(يشير إلى قوة واتجاه العلاقة).

Correlation – Case Study

- **Data Collection**:
- We collect data from 10 students.
- Record the number of hours each student studied and their exam scores.

Student	Hours Studied	Exam Score
A	2	50
B	3	55
C	4	60
D	5	65
E	6	70
…	….	….

- **Calculation**:
  - Use a formula or statistical software to calculate the Pearson correlation coefficient (r).
- **Result**:
- Let's say the calculated **r is 0.98**.
- **Interpretation**:
  - r = 0.98 indicates a strong positive correlation.
  - This means that as the number of hours studied increases, the exam score also increases.

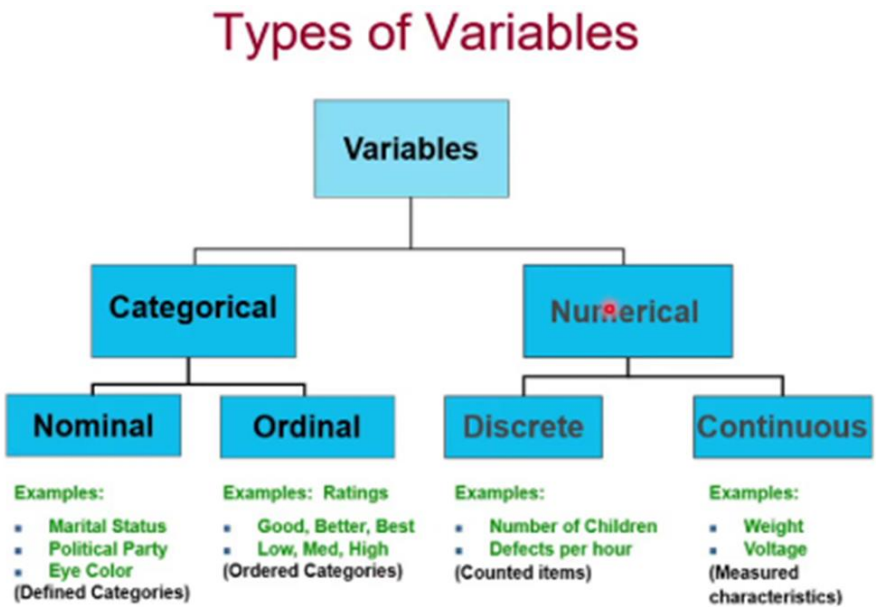
Pearson Correlation

- Types of Correlation Coefficient
  - Pearson
  - Spearman

Correlation coefficient	Type of relationship	Levels of measurement	Data distribution
Pearson's r	Linear	Two quantitative (interval or ratio) variables	Normal distribution
Spearman's rho	Non-linear	Two ordinal, interval or ratio variables	Any distribution

Correlation

- Apply Pearson Correlation:
  - Both variables are numeric(discrete or continuous)
  - The variables are normally distributed
    - Investigate through Shapiro-Wilkson Test
  - The data have no outliers
    - Check through histogram
- Apply Spearman Correlation:
  - Both variables are Ordinal
- Apply Chi-Square Test:
  - Both variables are Nominal



Correlation - Quiz

Find what type of correlation we can apply. The variables in square box are the variables to be correlated.

ID	Division	Quiz	Practical	Project	Final
H00247	EDU	84	67	69.6	67
H00258	HLT	74	68	67	73
H00269	ENG	88	86	75.6	74
H00224	EDU		71	65.8	71
H00269	EDU	22	65	66.8	78
H00277	ENG	40	68	76.4	68
H00277	HLT	71	77	64.6	70
H00293	BUS	50	74	60.6	78
H00247	CIS	61	70	66.4	82
H00149	APP	97	98	85.2	87
H00295	EDU	70	76	81	84
H00284	CIS	48	64	81.4	68
H00307	ENG	67	77	73	83
H00293	APP	38	89	64.6	77

Pearson correlation since it is numeric but have to be checked for normality

ID	Satisfaction_Level	Grade	Work_Experience
1	Satisfied	A	Advanced
2	Very Unsatisfied	A	Intermediate
3	Satisfied	A	Beginner
4	Very Satisfied	C	Advanced
5	Satisfied	B	Beginner
6	Unsatisfied	A	Beginner
7	Unsatisfied	C	Beginner
8	Unsatisfied	A	Intermediate
9	Neutral	B	Advanced
10	Neutral	B	None
11	Very Satisfied	A	None
12	Neutral	B	None
13	Neutral	C	Beginner
14	Very Unsatisfied	B	None

Spearman correlation since both variables are ordinal

	grade	sub_grade	loan_status	purpose
0	B	B2	Fully Paid	credit_Card
1	C	C4	Charged Off	credit_Card
2	C	C5	Fully Paid	credit_Card
3	C	C1	Fully Paid	credit_Card
4	B	B5	Fully Paid	credit_Card
5	A	A4	Fully Paid	credit_Card
6	C	C5	Fully Paid	debt_consolidation
7	E	E1	Fully Paid	credit_Card
8	F	F2	Charged Off	credit_Card
9	B	B5	Charged Off	credit_Card

Chi-square test since both variables are nominal

Pearson Correlation

Pearson correlation coefficient (r)	Correlation type	Interpretation	Example
Between 0 and 1	Positive correlation	When one variable changes, the other variable changes in the same direction.	Baby length & weight: The longer the baby, the heavier their weight.
0	No correlation	There is **no relationship** between the variables.	Car price & width of windshield wipers: The price of a car is not related to the width of its windshield wipers.
Between 0 and –1	Negative correlation	When one variable changes, the other variable changes in the **opposite direction**.	Altitude & air pressure: The higher the altitude, the lower the air pressure.

Pearson Correlation

- Check correlation between Project and Final marks

```python
# pearson correlation
import pandas as pd
from scipy.stats import pearsonr

# Load the dataset
file_path = 'Marks.csv'
df = pd.read_csv(file_path)

# Drop rows with missing values for Project and Final
df_clean = df.dropna(subset=['Project', 'Final'])

# Perform Pearson's Correlation
r, _ = pearsonr(df_clean['Project'], df_clean['Final'])

# Print only the Pearson correlation coefficient
print(f"Pearson's Correlation Coefficient: {r:.4f}")
```

```
Pearson's Correlation Coefficient: 0.2531
```

Magnitude of Correlation (r = 0.2531):

- The correlation coefficient r is **positive** and relatively small (close to 0).
- This indicates a **weak positive linear relationship** between **Project scores** and **Final scores**.
- A weak positive correlation means that as **Project scores** increase, **Final scores** tend to increase slightly, but the relationship is not strong.

Spearman Correlation

- Check correlation between Grade and Work experience

```python
# Spearman's Correlation
import pandas as pd
from scipy.stats import spearmanr

# Load the provided dataset
file_path = 'ordinal_dataset.csv'
df = pd.read_csv(file_path)

# Map ordinal variables to numerical ranks
grade_ranking = {'A': 1, 'B': 2, 'C': 3, 'D': 4, 'F': 5}
experience_ranking = {'None': 0, 'Beginner': 1, 'Intermediate': 2, 'Advanced': 3}

df['Grade_Rank'] = df['Grade'].map(grade_ranking)
df['Work_Experience_Rank'] = df['Work_Experience'].map(experience_ranking)

# Drop rows with missing values in relevant columns
df_clean = df.dropna(subset=['Grade_Rank', 'Work_Experience_Rank'])

# Perform Spearman's Correlation
r, _ = spearmanr(df_clean['Grade_Rank'], df_clean['Work_Experience_Rank'])

# Print only the correlation coefficient
print(f"Spearman's Correlation Coefficient: {r:.4f}")
```

```
Spearman's Correlation Coefficient: 0.0061
```

1. **Magnitude of Correlation (r = 0.0432):**
   - The value of r is very close to **0**, indicating a **very weak** (almost negligible) correlation between **Grade** and **Work Experience**.
   - A coefficient near zero suggests that changes in **Grade** are not associated with changes in **Work Experience**.
2. **Direction of Correlation:**
   - Since r is **positive**, it implies a slight positive relationship: as one variable increases, the other may tend to increase slightly. However, the magnitude is too small to consider this meaningful.

Conclusion:

- There is **no significant correlation** between **Grade** and **Work Experience**.
- The weak positive correlation is likely due to chance and does not indicate any meaningful relationship.

Correlation - Python

- To check correlation for all fields

```
#check correlation between the fields in the dataset
marks[1:30].corr(method="pearson")
```

	Quiz	Practical	Project	Final
Quiz	1.000000	0.368144	0.417109	0.364544
Practical	0.368144	1.000000	0.400262	0.341331
Project	0.417109	0.400262	1.000000	0.251626
Final	0.364544	0.341331	0.251626	1.000000

The figure shows that correlation between the exams were positively weakly corelated

Inference Statistics

Inference Statistics

- Inferential statistics is a branch of statistics that uses **statistical methods in sample** to **draw conclusions about a population**
- There are two main types of inferential statistics:
  - Hypothesis testing (testing the assumption)
    - used to test a hypothesis about a population parameter.
  - Confidence intervals
    - used to estimate a population parameter. based on sample data. (already covered in your pre-requisite)

Hypothesis Testing

Hypothesis Testing

- Why a hypothesis test is necessary alongside the correlation coefficient?
- **Correlation Coefficient Alone is Not Enough**
- The correlation coefficient (r) simply measures the magnitude and direction of the relationship:
  - Magnitude: How strong the relationship is (r value)
  - Direction: Positive or negative relationship.
- However:
  - If r is small (close to 0), it might be due to chance.
  - If r is large, it may still not be statistically significant in small samples.

Hypothesis Testing

- The hypothesis test helps us determine whether the observed correlation is real or due to random sampling variation.
  - **Null Hypothesis (H_0):** There is no correlation between the two variables ($\rho=0$).
  - **Alternative Hypothesis (H_1):** There is a correlation ($\rho\neq0$).
- The test produces a p-value:
- If p-value < 0.05, we reject the null hypothesis and conclude the correlation is statistically significant.
- If p-value > 0.05, we accept the null hypothesis and conclude the correlation is not significant.

Hypothesis Testing

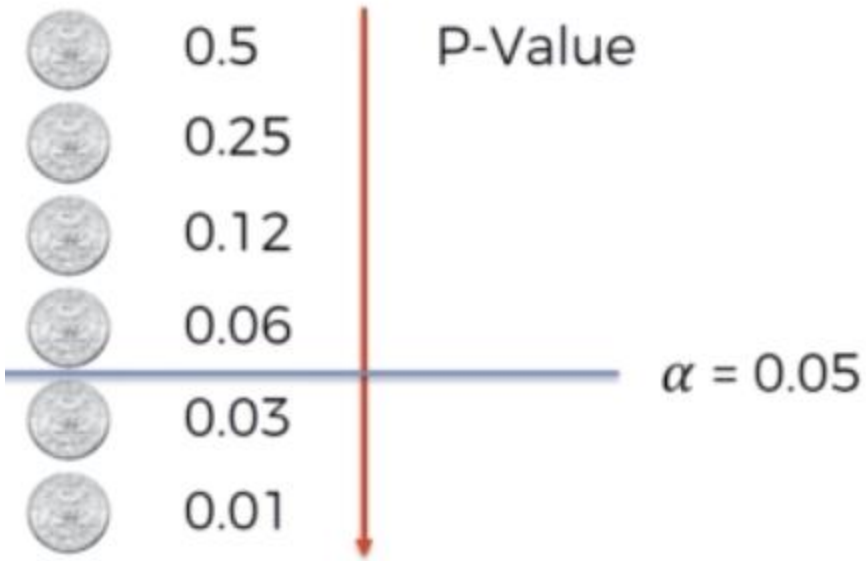
- **The Role of Sample Size**
  - In small sample sizes, even a large correlation coefficient may not be significant because there is more variability in smaller datasets.
  - Conversely, in large datasets, even a small correlation coefficient might be statistically significant.
  - Thus, the hypothesis test takes sample size into account to assess whether the correlation coefficient is meaningful.

Hypothesis Testing

- **P-value**:
- Measures the **statistical significance** of observed data.
- A lower p-value indicates a greater significance.
- A p-value ≤ 0.05 is statistically significant.

Hypothesis Testing– Case Study

- To determine whether a coin used for tossing is fair coin or not
- Null Hypothesis : It is a fair coin
- Alternative Hypothesis : It is not a fair coin
- Consider all the time when you toss the coin, you are getting tail. So the confidence level(p-value) will be decreasing about the fairness of the coin(null hypothesis)
- After the 4th toss (alpha(**α**)=0.05 – 95% confidence), we will reject the null hypothesis which means coin is not fair coin



Hypothesis Testing

- **Steps in performing Hypothesis Testing**:
  - ✓**Step 1**: Let assume the null hypothesis, alternate hypothesis, and the level of significance.
  - ✓**Step 2**: Calculate the P-value.
  - ✓**Step 3**: Conclude whether to reject the null hypothesis or not based on the P-value i.e.
    - ❖If **P-value < significance level**(0.05 for 95% confidence level), then **reject the null hypothesis**
    - ❖If **P-value >= significance level**, the **null hypothesis cannot be rejected**
  - ✓**Step 4:** State the conclusion.

Hypothesis Testing

- Check correlation between Project and Final marks
- From the earlier slide, we got the result as :

```
Pearson's Correlation Coefficient: 0.2531
```

- **Why Perform a Hypothesis Test?**

Determine Statistical Significance:

1. The **correlation coefficient (r)** tells us the strength and direction of the linear relationship between the variables (**Project Marks** and **Final Marks**).
2. However, to conclude whether this observed correlation is **statistically significant** (i.e., not due to random chance), we need a **hypothesis test**.

Hypothesis Testing – Case Study 1

- **Purpose**:
  - Test assumption that students who performed well on the project will score high in FWA.
- **Null Hypothesis (H_0):**
  - There is no significant correlation between Project marks and Final marks.
- **Alternative Hypothesis (H_1):**
  - There is a significant correlation between Project marks and Final marks.
- Apply **correlation**, and **hypothesis testing**.

Hypothesis Testing – Case Study 1

```python
#pearson correlation
from scipy.stats import pearsonr
import pandas as pd

# Load the dataset
df = pd.read_csv("Marks.csv")

# Extract Project and Final columns, dropping NaN values
project_scores = df['Project'].dropna()
final_scores = df['Final'].dropna()

# Ensure both series have equal length
min_length = min(len(project_scores), len(final_scores))
project_scores = project_scores[:min_length]
final_scores = final_scores[:min_length]

# Calculate Pearson correlation
r_value, p_value = pearsonr(project_scores, final_scores)
degrees_of_freedom = len(project_scores) - 2

# Print results
print(f"Pearson Correlation Coefficient (r): {r_value:.3f}")
print(f"P-Value: {p_value:.3f}")
print(f"Degrees of Freedom: {degrees_of_freedom}")

# Interpretation
if p_value < 0.05:
    print("\nConclusion: Since the p-value is less than 0.05, we reject the null hypothesis.")
    print("There is a statistically significant correlation between Project and Final marks.")
else:
    print("\nConclusion: Since the p-value is greater than 0.05, we accept the null hypothesis.")
    print("There is no statistically significant correlation between Project and Final marks.")
```

Output:

```
Pearson Correlation Coefficient (r): 0.253
P-Value: 0.244
Degrees of Freedom: 21

Conclusion: Since the p-value is greater than 0.05, we accept the null hypothesis.
There is no statistically significant correlation between Project and Final marks.
```

Why is Equal Length Important for Pearson Correlation?

Pearson correlation requires paired data points, meaning:

1. Each value in project_scores must correspond to a value in final_scores (one-to-one pairing).
2. If the lengths are unequal, it will result in misalignment and errors during computation.

Hypothesis Testing – Case Study 2

- Check correlation between Grade and Work experience
- From earlier slide, we got the result as :

```
Spearman's Correlation Coefficient: 0.0061
```

- **Why Perform a Hypothesis Test?**
  - **Correlation Coefficient Indicates Negligible Relationship**
    - The **Spearman's correlation coefficient** (r=0.0061) is extremely close to **0**, which already suggests that there is **no meaningful relationship** between **Grade** and **Work Experience**.
    - Since the correlation coefficient indicates a negligible or almost non-existent relationship, hypothesis testing is not required to check the significance.

Hypothesis Testing – Case Study 2

- However, spearman correlation is performed between project and final marks to understand the python code

- **Note: Only Pearson correlation should be performed between project and final marks since they are numeric**

Hypothesis Testing – Case Study 2

```python
#Spearman Rank Correlation
import pandas as pd
from scipy.stats import spearmanr

# Load the dataset
file_path = "Marks.csv"
df = pd.read_csv(file_path)

# Drop rows with NaN values for Project and Final
data = df[['Project', 'Final']].dropna()

# Perform Spearman's Correlation Test
r, p_value = spearmanr(data['Project'], data['Final'])
df_spearman = len(data) - 2

# Print the results
print("Spearman's Correlation Test Results:")
print(f"Spearman's Correlation Coefficient (r): {r:.4f}")
print(f"P-Value: {p_value:.4f}")
print(f"Degrees of Freedom: {df_spearman}")

# Interpretation
if p_value < 0.05:
    print("\nConclusion: Since the p-value is less than 0.05, we reject the null hypothesis.")
    print("There is a statistically significant correlation between Project and Final marks.")
else:
    print("\nConclusion: Since the p-value is greater than 0.05, we accept the null hypothesis.")
    print("There is no statistically significant correlation between Project and Final marks.")
```

Output:

```
Spearman's Correlation Test Results:
Spearman's Correlation Coefficient (r): 0.1700
P-Value: 0.4380
Degrees of Freedom: 21

Conclusion: Since the p-value is greater than 0.05, we accept the null hypothesis.
There is no statistically significant correlation between Project and Final marks.
```

"Degrees of Freedom: 21" indicates that the Spearman's correlation was calculated using 23 pairs of data points (Project marks and Final marks)

A higher df means a more reliable estimate of r, leading to a more accurate p-value.

Why subtract 2?
When performing a correlation analysis, you are estimating the relationship between two variables. The subtraction of 2 reflects the two variables being analyzed and removes the "degrees of freedom" used by the test.

Hypothesis Testing – Case Study 3

- **Purpose**: There is an assumption that loan status and card used to pay are interrelated.

  Is this assumption correct?

- **How To**: Hypothesis Test using Correlation test – Chi-square Test(since loan status and purpose are categorial variables)

	grade	sub_grade	loan_status	purpose
0	B	B2	Fully Paid	credit_Card
1	C	C4	Charged Off	credit_Card
2	C	C5	Fully Paid	credit_Card
3	C	C1	Fully Paid	credit_Card
4	B	B5	Fully Paid	credit_Card
5	A	A4	Fully Paid	credit_Card
6	C	C5	Fully Paid	debt_consolidation
7	E	E1	Fully Paid	credit_Card
8	F	F2	Charged Off	credit_Card
9	B	B5	Charged Off	credit_Card

Hypothesis Testing – Case Study 3

- Chi-Square Test–**Categorical (var1)Vs Categorical(var2)**
  - used to compare the observed frequencies of a categorical variable to the expected frequencies of that variable(يُستخدم لمقارنة الترددات الملاحظة لمتغير فئوي مع الترددات المتوقعة لذلك المتغير).
  - It is a non-parametric test, meaning that it does not make any assumptions about the distribution of the data.
  - Steps
    - Apply stratified sampling to extract var1 and var2
    - Comparing two variables[var1 Vs var2]
      - Apply chi-square – if both va1 and var2 are categorical variables
    - Hypothesis Test – var1 = var2 (Null) / var1 ≠ var2 (Alternative)
      - Check p-value
      - If p> 0.05, then null hypo rejected (var1≠ var2)
      - If p< 0.05, then null hypo accepted (var1=var2)

Hypothesis Testing – Case Study 3

- Chi-Square Test–**Categorical (var1)Vs Categorical(var2)**
  - Steps
    - Apply stratified sampling to extract var1 and var2
    - Comparing two variables[var1 Vs var2]
      - Apply chi-square – if both va1 and var2 are categorical variables
    - Hypothesis Test
      - Null Hypothesis (H_0): Loan status and purpose are independent (no association).
      - Alternative Hypothesis (H_1): Loan status and purpose are not independent (there is an association).
      - Check p-value
        - If $p > 0.05$, then null hypo rejected
        - If $p < 0.05$, then null hypo accepted

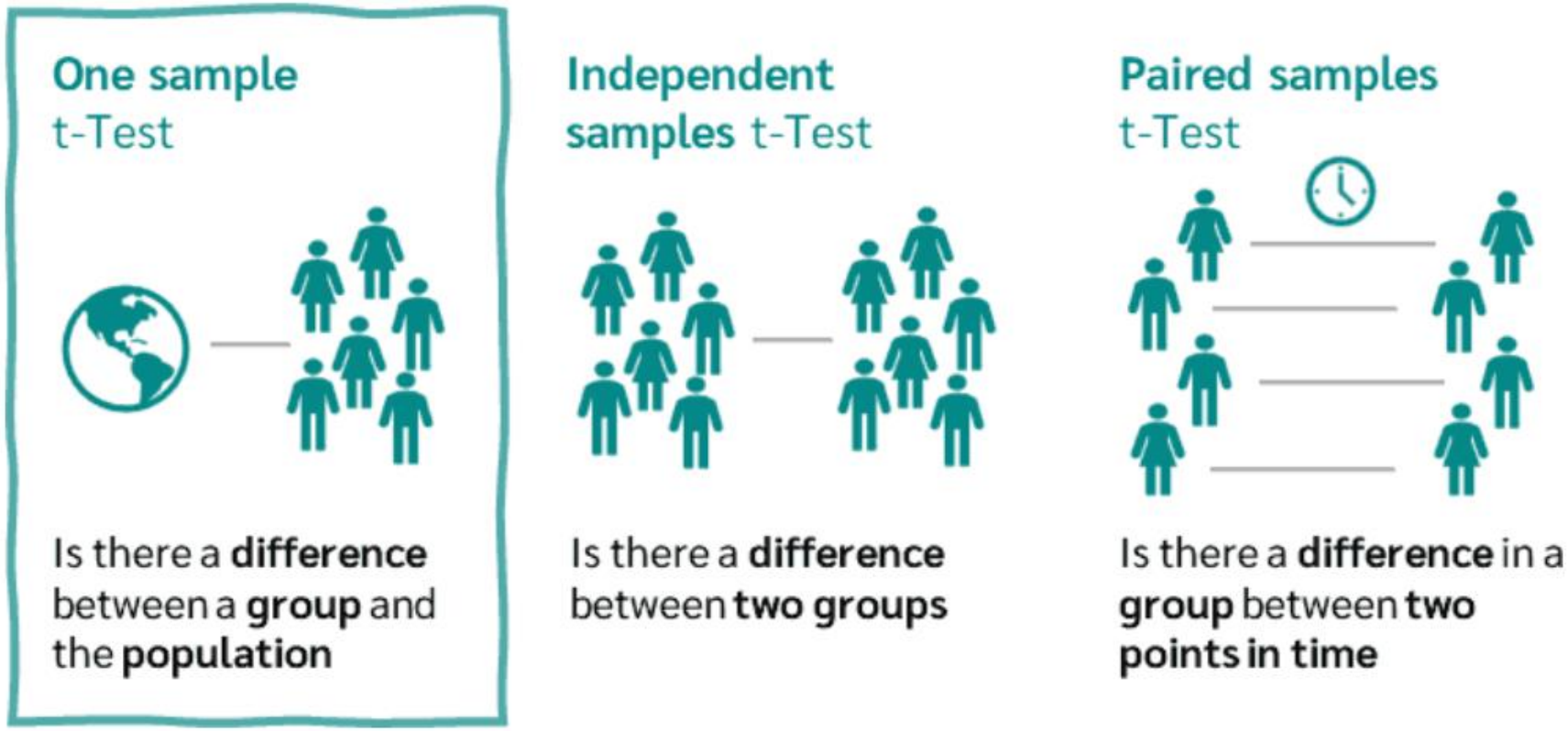
Hypothesis Testing – Chi-Square Test

```
#check chi-square for cross tab
from scipy.stats import chi2_contingency
r, p, dof, expected = chi2_contingency(contingency_data)
print('r=%.3f, p=%.3f' % (r, p))
if p > 0.05:
    print('independent categories')
else:
    print('dependent categories')
```

```
r=0.000, p=1.000
independent categories
```

P-Value ≥ 0.05: Accept the null hypothesis. This means there is no significant association between **loan status** and **purpose**.

Hypothesis Testing – One Sample T-test



Hypothesis Testing – One Sample T-test

- **One Sample T-test - Average(Sample) Vs Average(Population)**
  - used to compare the average of a single **sample** of data to a known or hypothesized **population** average.
  - For example, to test whether the average height of students in a particular school is different from the national average height for students of the same age.
- **Shapiro- Wilkson Method**
  - A statistical test used to assess the normality of a sample.
  - It is a non-parametric test, meaning that it does not make any assumptions about the distribution of the data.

Hypothesis Testing – Case Study 4

- **Purpose**: To find the average(mean) score of FWA by CIS students is same as average of all HCT students FWA marks.
- **How To**:
  - Apply stratified sampling to extract var1=avg(sample) and var2=avg(population)
  - Comparing two variables[var1 Vs var2]
    - Apply One sample T-test – if va1 and var2 are average of sample and population
  - Hypothesis Test – var1 = var2 (Null) / var1 ≠ var2 (Alternative)
    - Check p-value
    - If p< 0.05, then null hypothesis rejected (var1≠ var2)
    - If p> 0.05, then null hypothesis accepted (var1=var2)

Hypothesis Testing – Case Study 3

- **Steps**
  - Find the average of final marks of all the students(population)
  - Extract the final marks of CIS students(sample-stratified)
  - Check whether final marks are normally distributed using Shapiro-Wilkson method
  - Perform one sample T-test

Hypothesis Testing – Case Study 3

```
#Find the average of final marks of all the students
import pandas as pd
marks = pd.read_csv("Marks.csv")
Final=marks['Final']
print(Final.mean())
```

population

```
75.69565217391305
```

```
# Extract Final marks of CIS students
Final_CIS=marks[marks['Division']=='CIS']
Final_CIS
Final_CIS['Final']
```

sample

```
8     82
11    68
21    75
```

Hypothesis Testing – Case Study 3

```
#Check for normality
from scipy.stats import shapiro
r, p = shapiro(Final)
print('r=%.2f, p=%.30f' % (r, p))
if p > 0.05:
    print('Normal distribution')
else:
    print('Not a normal distribution')
```

```
r=0.94, p=0.202803328633308410644531250000
Normal distribution
```

```
#One Sample T-test
import scipy
scipy.stats.ttest_1samp(Final_CIS['Final'],75)
```

```
TtestResult(statistic=0.0, pvalue=1.0, df=2)
```

Since p value is greater then 0.05, **null hypothesis is accepted** meaning the average(mean) score of FWA by CIS students is same from average of all HCT students FWA marks

Hypothesis Testing – Case Study 3

Since p value is greater then 0.05, **null hypothesis is accepted** meaning the average(mean) score of FWA by CIS students is same from average of all HCT students FWA marks

The above can be rewritten as,

Fail to reject the null hypothesis: There is no significant difference between CIS and all HCT FWA scores.

Note: The second is preferred.

Project

11. Perform the hypothesis test to find the correlation (Pearson and Spearman for numerical variable and chi-square test for categorical variable) between the independent variable and the dependent variable. Note: If you have more than one independent variable, then chose any one of the independent variables.	1
12. Assess the performance of the dependent variable to know whether the sample is representative of the normal population by a one-sample t-test.	1

Conclusion

- **Understanding Correlation**:
- **Correlation Coefficient (r)** helps us measure the strength and direction of the relationship between two variables.
- **Types of Correlation:**
  - **Pearson Correlation** for linear relationships and normally distributed quantitative data.
  - **Spearman's Correlation** for non-linear relationships and ordinal data.
- Practical applications in Python demonstrated how to calculate and interpret correlation coefficients using sample data.

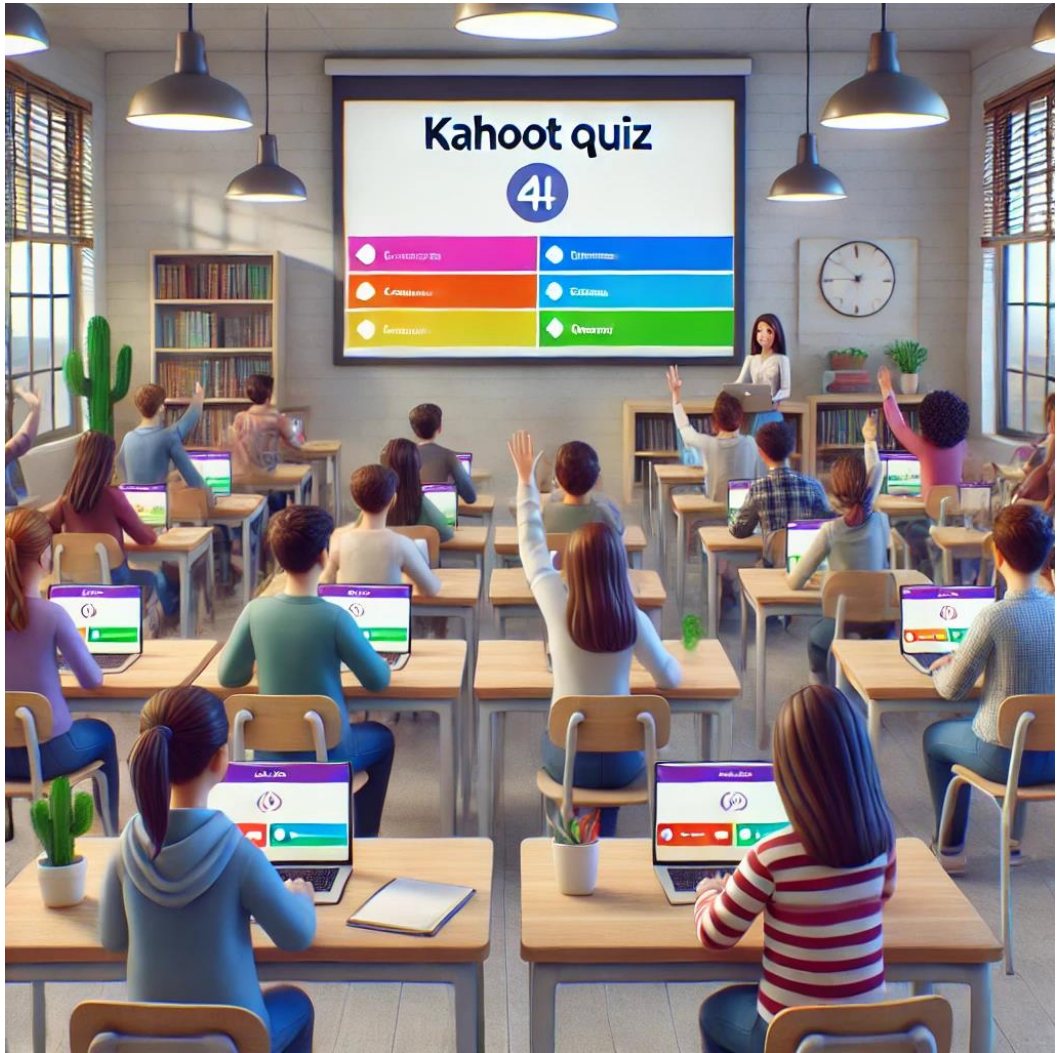
Conclusion

- **Hypothesis Testing**:
- **Hypothesis Testing** allows us to make inferences about a population based on sample data.
- The process involves:
  - Formulating the null (H0) and alternative (H1) hypotheses.
  - Calculating the p-value to determine statistical significance.
  - Making a decision based on the p-value relative to the significance level.
- Various hypothesis tests, including the **Chi-Square Test** and **One Sample T-test**, were applied to real-world scenarios to illustrate their use in data analysis.

Conclusion

- **Practical Application**:
- Through Python programming, we performed correlation analysis and hypothesis testing on sample datasets.
- We conducted multiple case studies to understand the relationship between variables such as project marks and final exam scores, loan status and card type, and average scores of different student groups.
- The results from these case studies emphasized the importance of statistical inference in drawing meaningful conclusions from data.

Kahoot Time!



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Thank You